

Final Project Report : Cryptocurrency Volatility Prediction System

1.High-Level Design (HLD):

1.1 Objective The objective of this project is to design and deploy a machine learning-based cryptocurrency volatility prediction system. The system aims to forecast 7-day rolling volatility for multiple cryptocurrencies to support risk management and informed decision-making by quantifying market "noisiness".

1.2 System Overview The system processes historical cryptocurrency market data through a structured pipeline:

- **Data Ingestion:** Loading historical data from a multi-cryptocurrency CSV dataset.
- **Preprocessing:** Cleaning timestamps and handling missing or infinite values.
- **Feature Engineering:** Extracting volatility and liquidity indicators to create predictive "signals".
- **Modeling:** Training a Random Forest Regressor to learn patterns in market behavior.
- **Deployment:** Creating an interactive web interface using Streamlit for real-time visualization.

1.3 Technology Stack

- **Programming Language:** Python
- **Data Processing:** Pandas, NumPy
- **Machine Learning:** Scikit-learn (Random Forest Regressor)
- **UI & Deployment:** Streamlit

2. Low-Level Design (LLD)

2.1 Data Preprocessing

- **Cleaning:** Managed timestamps and handled missing values using row elimination.
- **Standardization:** Applied StandardScaler to normalize features so that large-scale values like Market Cap do not overpower small-scale values like daily returns.

2.2 Feature Engineering The model utilizes six core engineered features:

1. **Daily Return:** Percentage price change since the previous day.
2. **HL Spread:** Normalized difference between the high and low price to capture intraday panic.

3. **OC Change:** Measures daily market momentum by comparing Open vs. Close prices.
4. **Volume:** Total dollar amount traded in 24 hours.
5. **Market Cap:** Total market value representing asset stability.
6. **Liquidity Ratio:** (Volume / Market Cap) showing market depth and ease of trading.

2.3 The Target Variable The model predicts the 7-day rolling standard deviation of daily returns, which represents the mathematical definition of market volatility.

3. Exploratory Data Analysis (EDA) The analysis phase confirmed that volatility follows detectable patterns:

- **Volatility Clustering:** Periods of high volatility often occur in groups following sharp price movements.
- **Correlation:** A strong positive correlation was found between the HL Spread and 7-day volatility.

4. Observations & Analysis

4.1 Key Observations

- **Market Patterns:** Volatility is not random; high volatility periods tend to follow one another, often triggered by sharp price movements.
- **Signal Reliability:** The High-Low Spread (HL Spread) showed the strongest positive correlation with 7-day volatility, confirming that intraday price instability is a highly reliable predictor.
- **Asset Sensitivity:** Smaller cryptocurrencies with lower Market Caps demonstrated significantly higher sensitivity to volume changes than established assets.

4.2 Technical Analysis

- **Liquidity Insights:** Analysis showed that lower Liquidity Ratios (Volume/Market Cap) often signal a higher susceptibility to sudden, extreme price swings.
- **Algorithm Performance:** The Random Forest Regressor effectively captured non-linear relationships in financial data, accurately predicting risk levels during varying market conditions.

5. Deployment & Outcome

5.1 Streamlit Interface

The final product is a functional web application:

- **Interactive Sidebar:** Allows users to manually adjust market metrics via sliders.
- **Real-time Inference:** The app loads serialized `.pkl` files to provide instant forecasts.

5.2 Final Results

- **Risk Quantification:** Successfully transformed "market fear" into a predictable mathematical value.
- **Technical Integration:** Managed large model files (over 25MB) using **Git LFS** for professional GitHub hosting.
- **Decision Support:** Provides a clear indicator for investors to identify high-risk trading periods.

6. Conclusion

The Cryptocurrency Volatility Prediction System successfully achieved its goal of creating an end-to-end risk assessment tool. By integrating data science with an interactive UI, the project proves that while crypto prices are unpredictable, their volatility follows logical patterns that can be modeled for better risk management.